CSCI426 · WEB PROGRAMMING

HealthTrack

A personal wellness tracker for everyday movement, meals, and hydration.

Project Report

Course	CSCI426 — Web Programming Advanced
Project theme	HealthTrack
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01 · ABSTRACT

Project overview

HealthTrack is a responsive React frontend for recording small, everyday wellness activities. Users can review a daily overview, add movement and meal entries, record water intake, and browse their activity log. The interface includes five pages: Overview, Daily log, Features, About, and Contact.

The project demonstrates component-based UI development, state updates in response to user actions, controlled forms, list rendering and filtering, and browser storage. The application runs in the browser and stores its log as JSON in local storage. It does not include a server, database, user accounts, or medical functionality.

> **Project purpose**
> Provide a simple place to notice and organize personal wellness habits without presenting the information as medical advice.

Objectives

- Build a multi-page user interface with reusable React components.
- Support responsive layouts for desktop and mobile screens.
- Let users add and review daily movement, meals, and hydration.
- Keep the activity log in the current browser using JSON and local storage.

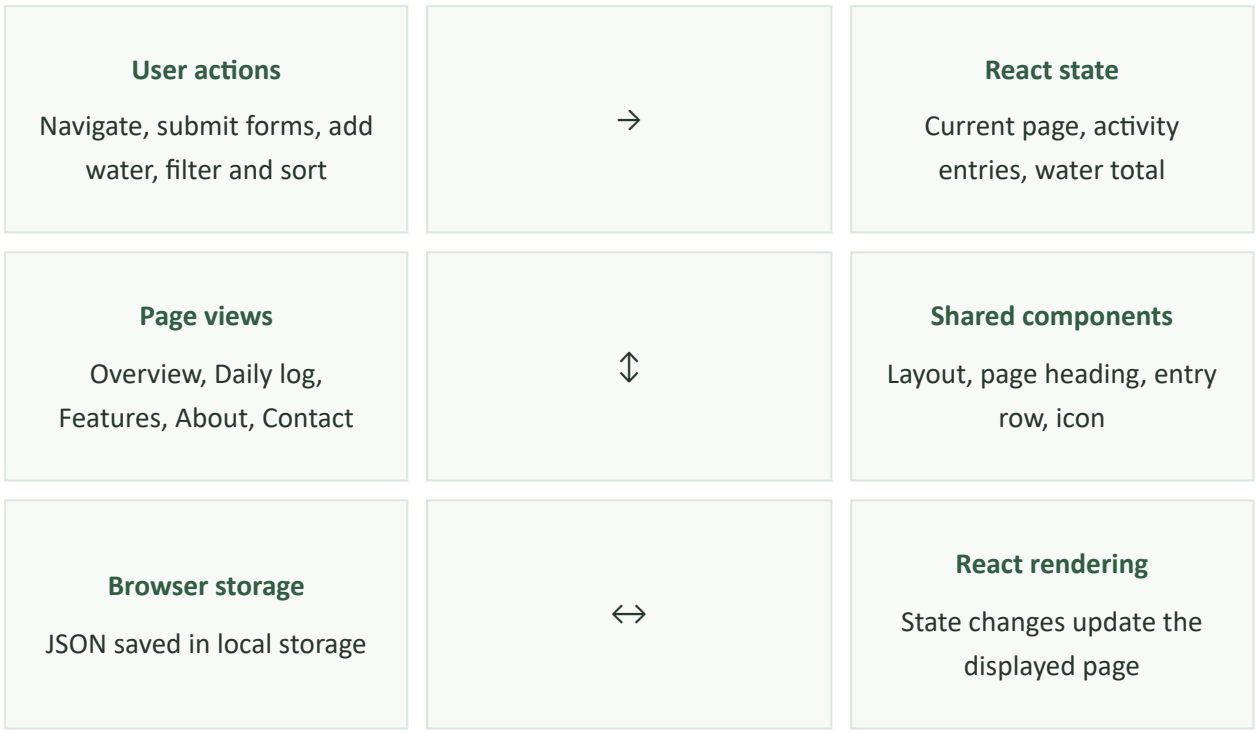
Scope

The Daily log is the dynamic page. Users can add movement or meal entries, add a glass of water, filter entries by type, and change their display order. The Contact page shows a local confirmation and does not send messages to a server.

02 · SYSTEM DESIGN

Frontend design

HealthTrack is a client-side React application. The main app keeps shared activity data and the selected page. Page components receive data and event callbacks through props. Reusable components provide the navigation layout, page headings, icons, and activity rows.



Data flow

A form event calls a handler in the app. The handler updates React state with the new entry. React renders the updated activity list, and an effect saves the latest state to local storage as JSON. When the app starts, it reads and parses saved data from the browser.

Main information

Information	Purpose
Water total	Tracks the amount of water added during the day.
Activity entries	Keeps movement, meal, and water records in a list.
Entry details	Stores the type, title, and optional note for each record.

> **Frontend boundary**
> Data stays in the current browser. The app has no backend or database connection.

03 · TECHNOLOGIES

Tools and concepts

Technology	Use in HealthTrack
React	Function components, props, state, and page composition.
JavaScript / ECMAScript	Arrow functions, variables, arrays, event handlers, and data updates.
JSX	HTML-like markup returned by React components.
CSS	Page layout, navigation, forms, and responsive behavior.
JSON and local storage	Convert activity data to a string and save it in the current browser.
Node.js and npm	Run the React app scripts and manage the project packages.

React concepts applied

- **Components and props:** page views are composed from reusable interface components; the parent passes data and callbacks to child components.
- **State and events:** state holds the selected page, form values, and activity data; click, change, and submit handlers respond to user actions.
- **Effects:** an effect saves changed activity data to local storage.
- **Lists:** array methods render entries and filter or sort the daily log.
- **Forms:** controlled form fields read and update React state.

Limitations

The app is a frontend demonstration. It does not synchronize data between devices, create user accounts, send contact messages, or provide medical recommendations.

04 · KEY CODE SNIPPETS

Implementation examples

1. Read and save JSON data

Saved browser data is restored when the app starts. Updated state is saved as JSON.

```
const STORAGE_KEY = 'healthtrack-data-v1';

function readSavedData() {
  try {
    const saved = localStorage.getItem(STORAGE_KEY);
    if (!saved) return initialData;
    const parsed = JSON.parse(saved);
    return {
      water: parsed.water || 0,
      entries: parsed.entries || [],
    };
  } catch {
    return initialData;
  }
}

useEffect(() => {
  localStorage.setItem(STORAGE_KEY, JSON.stringify(data));
}, [data]);
```

2. Add an item with React state

The event handler updates the entries array through the state setter.

```
const addEntry = (event, form) => {
  event.preventDefault();
  const entry = {
    type: form.type,
    title: form.title,
    detail: form.detail,
  };
  setData((current) => {
    const nextEntry = {
      ...entry,
      id: getNextId(current.entries),
    };
    return {
      ...current,
      entries: [nextEntry, ...current.entries],
    };
  });
};
```

3. Filter, sort, and render entries

The daily log uses array methods to select the records and render a reusable row for each one.

```
const visibleEntries = data.entries
  .filter((entry) => filter === 'all' || entry.type === filter)
  .sort((a, b) => sortOrder === 'newest'
    ? b.id - a.id
    : a.id - b.id);

{visibleEntries.map((entry) => (
  <EntryRow entry={entry} key={entry.id} />
))}
```

HealthTrack · CSCI426 Web Programming Advanced. This frontend is for general wellness organization and is not a medical service.